

Long Division

A Year 4 Maths Guide

Long division is a way of dividing big numbers into equal groups, one step at a time. We use the letters **D**ivide, **M**ultiply, **S**ubtract, and **B**ring down to remember the order — some children like to remember it as “**Dad, Mum, Sister, Brother**”!

The 4 Steps

1	Divide	How many times does the divisor fit into the digit?
2	Multiply	Multiply the divisor by that number.
3	Subtract	Take away the answer from the digit.
4	Bring down	Bring down the next digit and repeat.

Worked Example: $96 \div 4$

We want to share 96 into 4 equal groups. We work through the tens first, then the units.

Step	What we do	Working
Divide	How many 4s go into 9 (tens)?	4 goes into 9, 2 times (2 tens)
Multiply	$2 \times 4 = 8$	Write 8 under the 9
Subtract	$9 - 8 = 1$	1 ten left over
Bring down	Bring down the 6 (units)	Now we have 16
Divide	How many 4s go into 16?	4 goes into 16, 4 times
Multiply	$4 \times 4 = 16$	Write 16 under the 16
Subtract	$16 - 16 = 0$	Nothing left over — done!

So $96 \div 4 = 24$, with no remainder.

What if there's a remainder?

Sometimes numbers don't divide exactly. For example, $97 \div 4$ works the same way, but after the last subtraction we are left with 1 that cannot be shared into groups of 4 anymore. We write this as **24 remainder 1** (or 24 r1).

Top Tips

- Always start dividing from the left-hand digit (the biggest place value).
- If the divisor doesn't fit into a digit, bring down the next digit and try again.
- Line up your numbers neatly in columns — it helps avoid mistakes!

- Check your answer by multiplying it back: quotient $\times$ divisor (+ remainder) should equal the original number.
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Try It Yourself

Have a go at these using the Divide, Multiply, Subtract, Bring Down method:

$84 \div 4$	$78 \div 3$	$95 \div 5$	$63 \div 3$
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